

t-SNE Project Warmup Exercise

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1 Setup

This document is dedicated to the reconstruction of [1] with low-dimension kernel g replaced by a more general kernel. This would serve as a great way to understand the material and help with the completion of the original project. For the sake of simplicity, we will use the exact same notation as in [1] and skip all the definitions. Recall in [1], g is defined as

$$g(y, y') = \frac{1}{1 + \|y - y'\|^2} \quad (1.1)$$

With all these being said, we will first start by giving our general low dimension kernel g some general assumptions, and modify it as we move on. We first require that $g(y, y') = k(\|y - y'\|)$ such that:

1. $k : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is decreasing.
2. k satisfies

$$\iint_{(0, \infty)^2} k(x, x') dx dx' > \infty \quad (1.2)$$

We will kick off this warmup by listing all the important lemmas and results in order, and prove our own versions when the need arises. The final goal is to prove a same set of results as in [1] with the y kernel replaced. Recall that

$$\begin{aligned} F_{\rho, \mu}(x, \sigma) = & - \int \frac{\|x - x'\|^2}{2\sigma^2} \frac{\exp(-\|x - x'\|^2 / 2\sigma^2)}{\int \exp(-\|x - x'\|^2 / 2\sigma^2) d\mu(x')} d\mu(x') \\ & - \log \left(\int \exp(-\|x - x'\|^2 / 2\sigma^2) \right) + \log \rho. \end{aligned} \quad (1.3)$$

Proposition 1. *If μ has a C^1 density f such that both f and $\|\nabla f\|$ are bounded, then the equation $F_{\rho, \mu}(x, \sigma) = 0$ has a unique solution $\sigma_{\rho, \mu}^*(x)$ for all $\rho \in (0, 1)$ and $x \in \mathbb{R}^d$. Moreover, $x \mapsto \sigma_{\rho, \mu}^*(x)$ is smooth.*

Proof. This proposition is about the kernel of x . The proof stays unchanged from [1]. □

The next couple of lemmas are general results in section 3 of [1]. We will omit the proof because they are not directly discussing the quantities in the t-sne algorithm. Recall that X or the corresponding probability measure μ is called sub-Gaussian if $\mathbb{P}(X > t) \leq 2 \exp(-Ct^2)$ for some positive constant C .

Lemma 1. Let ν be sub-Gaussian with second moment equal to 1 and suppose $X \in \mathbb{R}^d$ has i.i.d. entries drawn from ν . Then there exists constant C, c_1 such that

$$\mathbb{P}(\|X\| \geq \sqrt{d} + t) \leq Ce^{-c_1 t^2}. \quad (1.4)$$

Lemma 2. Let μ be sub-Gaussian and c_1 be the constant from lemma 1. Let $X_1, \dots, X_n \in \mathbb{R}^d$ have i.i.d. entries drawn from μ . Then there exists a constant C such that

$$\mathbb{P}\left(\max_{i=1, \dots, n} \|X_i\| \geq \sqrt{\frac{\log n}{c_1}} + \sqrt{d} + t\right) \leq Ce^{-c_1 t^2}. \quad (1.5)$$

Lemma 3. Let $\delta > 0$ be constant, $K_n = [\delta, \sqrt{n}]$ and $h : \mathbb{R}^d \times \mathbb{R}^d \times \mathbb{R} \rightarrow \mathbb{R}$. Suppose the following hold:

1. μ is sub-Gaussian;
2. $h(x, x', \sigma)$ and $\nabla_x h(x, x', \sigma)$ are bounded for all x, x' and all $\delta \in K_n$;
3. $h(x, x', \sigma) \rightarrow 0$ as $\|x - x'\|/\sigma \rightarrow \infty$.

Let $X_1, \dots, X_n$ be i.i.d. random variables drawn from μ . Let $\mu_n = (1/n) \sum_i \delta_{X_i}$. Then for any $\epsilon > 0$ with probability at least $1 - e^{-c\epsilon^2 n}$ we have

$$\sup_{\sigma \in K_n, x} \left| \int h(x, x', \sigma) d\mu_n(x) - \int h(x, x', \sigma) d\mu(x') \right| \leq \epsilon + O\left(\sqrt{\frac{\log n}{n}}\right). \quad (1.6)$$

Define

$$f_\sigma(x, x') = \exp(-\|x - x'\|^2 / 2\sigma^2), \quad g_\sigma = \frac{\|x - x'\|^2}{2\sigma^2} \exp(-\|x - x'\|^2 / 2\sigma^2). \quad (1.7)$$

The next two lemmas use similar techniques to prove.

Lemma 4. Let c_1 be the decay constant in lemma 1 and let $\delta > 0$. Then almost surely in $X_1, \dots, X_n$,

$$\sup_{\substack{\sigma \in [\delta, \sqrt{n}] \\ i=1, \dots, n}} \left| \frac{\int g_\sigma(X_i, x') d\mu_n(x')}{\int f_\sigma(X_i, x') d\mu_n(x')} - \frac{\int g_\sigma(X_i, x') d\mu(x')}{\int f_\sigma(X_i, x') d\mu(x')} \right| = O(n^{8/c_1 \delta^2 - 1/2} \sqrt{\log n}). \quad (1.8)$$

Lemma 5. Let c_1 be the decay constant in lemma 1 and let $\delta > 0$ be such that $c_1 \delta^2 > 8$. Then almost surely in $X_1, \dots, X_n$,

$$\sup_{\substack{\sigma \in [\delta, \sqrt{n}] \\ i=1, \dots, n}} \left| \log \left(\int f_\sigma(X_i, x') d\mu_n(x') \right) - \log \left(\int f_\sigma(X_i, x') d\mu(x') \right) \right| = O(n^{4/c_1 \delta^2 - 1/2} \sqrt{\log n}) \quad (1.9)$$

For this pair of lemmas, [1] provides the proof to the first lemma. We will provide the second for the sake of exercise.

Proof of lemma 5. First, notice that for arbitrary $x > 0$, the following elementary inequality holds:

$$\log(x) \leq x - 1. \quad (1.10)$$

Therefore, the lemma can be proved if we are able to show that

$$\begin{aligned} \left| \frac{\int f_\sigma(X_i, x') d\mu_n(x')}{\int f_\sigma(X_i, x') d\mu(x')} - 1 \right| &= \left| \frac{\int f_\sigma(X_i, x') d\mu_n(x') - \int f_\sigma(X_i, x') d\mu(x')}{\int f_\sigma(X_i, x') d\mu(x')} \right| \\ &= O(n^{4/c_1 \delta^2 - 1/2} \sqrt{\log n}). \end{aligned} \quad (1.11)$$

We can achieve that by bounding the numerator and the denominator of the previous formula. On one hand, consider set

$$E'_1 = \left\{ \max_{i=1, \dots, n} \|X_i\| \geq \sqrt{\alpha + \frac{1}{c_1}} \sqrt{\log n} + \sqrt{d} \right\} \quad (1.12)$$

Setting $t = \sqrt{\alpha \log n}$ in lemma 2, we have $\mathbb{P}(E'_1) \leq Cn^{-c_1\alpha}$. To absorb the $\sqrt{d}$ term, we can find $\epsilon_n \rightarrow 0$ such that

$$E_1 = \left\{ \max_{i=1, \dots, n} \|X_i\| \geq \sqrt{\alpha + \frac{1}{c_1} + \epsilon_n} \sqrt{\log n} \right\} \quad (1.13)$$

with $\mathbb{P}(E_1) \leq Cn^{-c_1\alpha}$. Because the maximum of norm of each X_i is restricted, on E_1^c we have for all i, j , that

$$\|X_i - X_j\|^2 \leq 4 \left(\alpha + \frac{1}{c_1} + \epsilon_n \right) \log n =: 4\gamma \log n. \quad (1.14)$$

Then for $\sigma \in [\delta, \sqrt{n}]$, on E_1^c for all i we have

$$\int f_\sigma(X_i, x') d\mu(x') \geq (1 - Cn^{-c_1\alpha}) \exp\left(-\frac{2\gamma \log n}{\delta^2}\right) \geq \frac{1}{2} \exp\left(-\frac{2\gamma \log n}{\delta^2}\right) = \frac{1}{2} n^{-2\gamma/\delta^2}. \quad (1.15)$$

This helps us bound the denominator. On the other hand, by lemma 3, we are able to find set E_2 with $\mathbb{P}(E_2) \leq e^{-c\epsilon^2 n}$, such that, on E_2^c ,

$$\sup_{\substack{\sigma \in [\delta, \sqrt{n}] \\ i=1, \dots, n}} \left| \int f_\sigma(X_i, x') d\mu_n(x') - \int f_\sigma(X_i, x') d\mu(x') \right| \leq \epsilon + O\left(\frac{\sqrt{\log n}}{\sqrt{n}}\right). \quad (1.16)$$

Therefore, we can combine (1.15) and (1.16) and conclude that on $E_1^c \cap E_2^c$,

$$\left| \frac{\int f_\sigma(X_i, x') d\mu_n(x') - \int f_\sigma(X_i, x') d\mu(x')}{\int f_\sigma(X_i, x') d\mu(x')} \right| \leq C\epsilon n^{2\gamma/\delta^2} + O(\sqrt{\log n} \cdot n^{-1/2+2\gamma/\delta^2}). \quad (1.17)$$

In order for the RHS to converge to zero, we need

$$\epsilon n^{2\gamma/\delta^2} \rightarrow 0, \quad 1/2 - 2\gamma/\delta^2 > 0. \quad (1.18)$$

Moreover, we have $\mathbb{P}(E_1 \cup E_2) \leq Cn^{-c_1\alpha} + e^{-c\epsilon^2 n}$. Set $\epsilon = n^{-\eta}$ for some $\eta > 0$. In order to get almost sure convergence using Borel-Cantelli, we also need

$$c_1\alpha > 1, \quad 1 - 2\eta > 0. \quad (1.19)$$

Hence for the entire lemma to work we need the following four inequalities to hold:

$$\frac{2\gamma}{\delta^2} - \eta < 0, \quad \frac{2\gamma}{\delta^2} - 1/2 < 0, \quad c_1\alpha > 1, \quad 1 - 2\eta > 0. \quad (1.20)$$

The last inequality implies that $\eta < 1/2$ so the first inequality implies the second. Now, recall that

$$\gamma := \alpha + \frac{1}{c_1} + \epsilon_n. \quad (1.21)$$

The collection of inequalities is equivalent to

$$2 \left(\alpha + \frac{1}{c_1} + \epsilon_n \right) < \eta \delta^2, \quad c_1\alpha > 1, \quad 0 < \eta < 1/2. \quad (1.22)$$

Here, we are free to choose α and η , but c_1 and δ are fixed. Now let

$$\alpha = \frac{1}{c_1} + \epsilon, \quad \eta = \frac{1}{2} - \epsilon. \quad (1.23)$$

Then the second and third inequalities in (1.22) are satisfied. The first is equivalent to

$$\frac{4}{c_1 \delta^2} + \epsilon < \frac{1}{2} - \epsilon \quad (1.24)$$

as $\epsilon_n \rightarrow 0$ as $n \rightarrow \infty$. In addition, since ϵ is arbitrary, we have again, equivalently,

$$\frac{4}{c_1 \delta^2} - \frac{1}{2} < 0. \quad (1.25)$$

When this is satisfied, we have:

$$\sup_{\substack{\sigma \in [\delta, \sqrt{n}] \\ i=1, \dots, n}} \left| \int f_\sigma(X_i, x') d\mu_n(x') - \int f_\sigma(X_i, x') d\mu(x') \right| = O(n^{4/c_1 \delta^2 - 1/2} \sqrt{\log n}). \quad (1.26)$$

The proof is thus finished. $\square$

Then combining the previous result we have an important convergence:

Proposition 2. *The μ be sub-Gaussian and c_1 be the constant from lemma 1. Let $X_1, \dots, X_n$ be i.i.d. random variables with probability measure μ and $\mu_n = \frac{1}{n} \sum \delta_{X_i}$ be the empirical measure. Let $\delta > 0$ be a constant such that $c_1 \delta^2 > 16$. Then, there exists $\epsilon > 0$, such that almost surely in $X_1, \dots, X_n$:*

$$\sup_{\substack{\sigma \in [\delta, \sqrt{n}] \\ i=1, \dots, n}} |F_{\mu_n}(X_i, \sigma) - F_\mu(X_i, \sigma)| = O(n^{-\epsilon}). \quad (1.27)$$

Let us summarize the important results upon this point, which will be used later:

There exists $\gamma = 2/c_1 + \epsilon$ such that the following holds almost surely

- (1) $\max_{i=1, \dots, n} \|X_i\| \leq \sqrt{\gamma \log n}$;
- (2) $\max_{i,j} \|X_i - X_j\|^2 \leq 4\gamma \log n$;
- (3) For $\sigma \in [\delta, \sqrt{n}]$,

$$\int \exp(-\|X_i - x'\|^2 / 2\sigma^2) d\mu_n(x), \int \exp(-\|X_i - x'\|^2 / 2\sigma^2) d\mu(x') \geq C n^{-2\gamma/\sigma^2}. \quad (1.28)$$

Next couple of lemmas in [1] are about lower bound on $\sigma_{\rho, \mu}^*$. Recall that $\sigma_{\rho, \mu}^*$ is the unique solution of the equation $F_{\rho, \mu}(x, \sigma_{\rho, \mu}^*) = 0$. For simplicity of notation, we write $\sigma^*(x) = \sigma_{\rho, \mu}^*(x)$ and $\sigma_n^*(x) = \sigma_{\rho, \mu_n}^*(x)$.

Lemma 6. *If μ satisfies the condition in lemma 1 of [1], then $\sigma^*(x)$ and $\sigma_n^*(x)$ are uniformly bounded from below in both i and n almost surely.*

Proposition 3. *Suppose μ has compact support and C^1 density. Then almost surely, as $n \rightarrow \infty$,*

$$\sup_{i=1, \dots, n} |\sigma_n^*(X_i) - \sigma^*(X_i)| \rightarrow 0. \quad (1.29)$$

Proof. By [1], we can quickly reach to the stage where

$$|\sigma_n^*(x) - \sigma^*(x)| = O(|F_{\mu_n}(x, \sigma_n^*(x)) - F_{\mu}(x, \sigma_n^*(x))|) \rightarrow 0. \quad (1.30)$$

Moreover, we know that μ has compact support and $x \mapsto \sigma^*(x)$ is smooth, $\sigma^*(X_i)$ is bounded above and below, meaning that $\sigma_n^*(X_i)$ is bounded uniformly from above and below as well. This completes the proof in the original paper. $\square$

References

- [1] Antonio Auffinger and Daniel Fletcher. Equilibrium distributions for t-distributed stochastic neighbour embedding, 2023, 2304.03727. URL <https://arxiv.org/abs/2304.03727>.
- [2] Aditya Ravuri and Neil D. Lawrence. Towards one model for classical dimensionality reduction: A probabilistic perspective on umap and t-sne, 2024, 2405.17412. URL <https://arxiv.org/abs/2405.17412>.
- [3] Pavlin G. Poličar and Blaž Zupan. Visualizing high-dimensional temporal data using direction-aware t-sne, 2024, 2403.19040. URL <https://arxiv.org/abs/2403.19040>.
- [4] Laurens van der Maaten and Geoffrey Hinton. Visualizing data using t-sne. *Journal of Machine Learning Research*, 9(2579-2605), 2008.
- [5] Stéphane Boucheron, Gábor Lugosi, and Pascal Massart. *Concentration Inequalities: A Nonasymptotic Theory of Independence*. Oxford University Press, 2013.